

Machine Learning

Lecture 8: Bias-Variance tradeoff, SHAP

Iurii Efimov



Outline

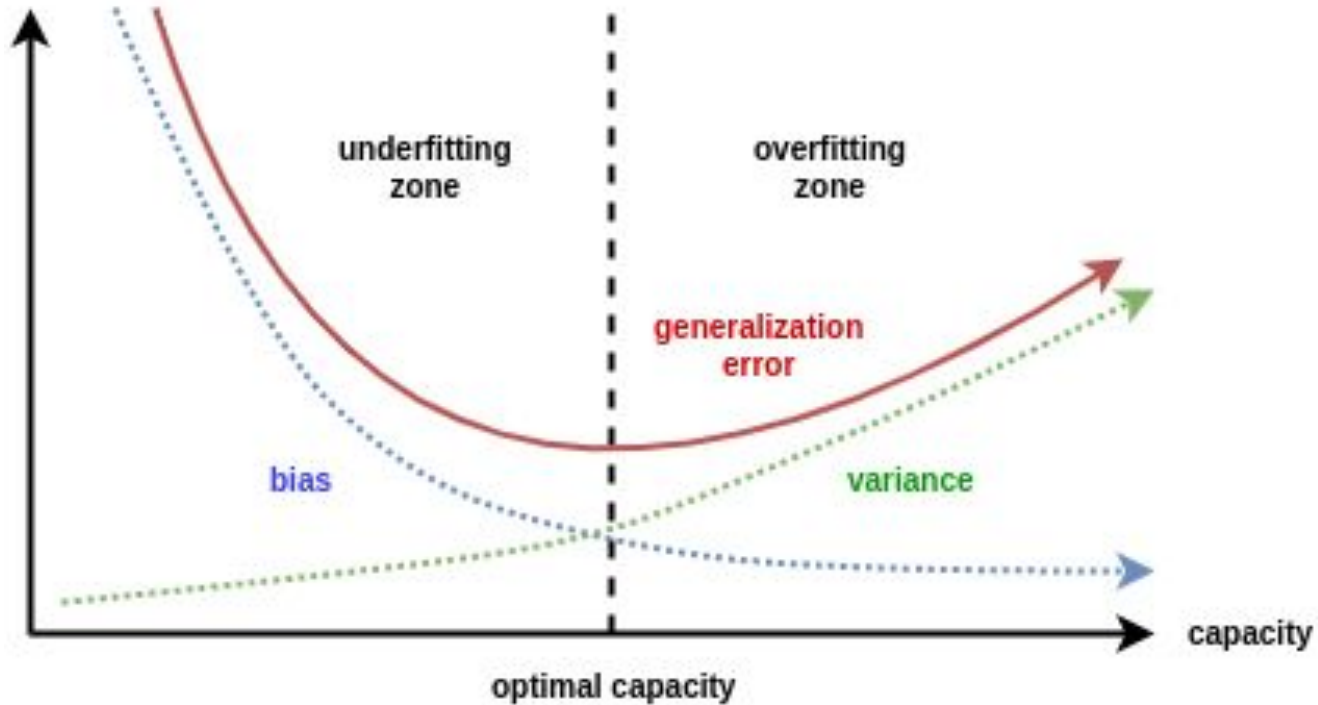
- Bias-Variance Tradeoff
- SHAP Values

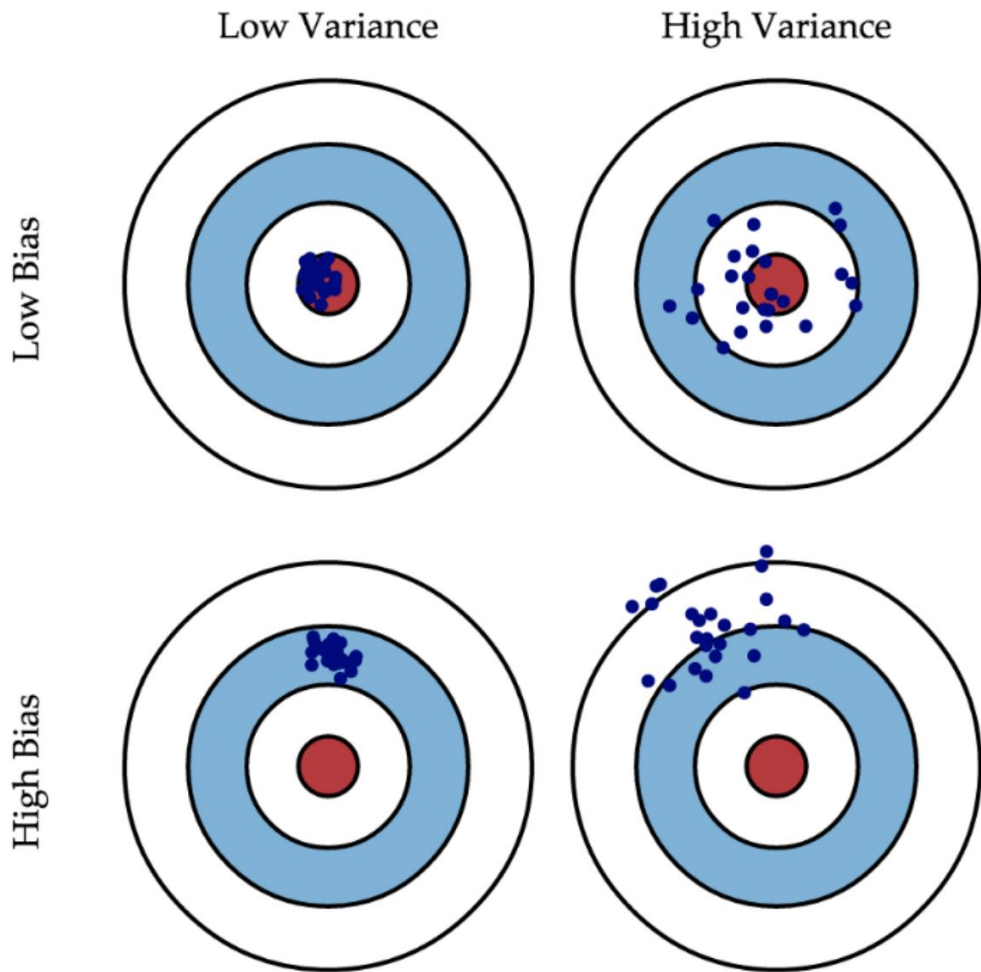
Bias-Variance tradeoff

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01

Bias-variance tradeoff







$$L(\mu) = \underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{noise}} + \underbrace{\mathbb{E}_x \left[(\mathbb{E}_X [\mu(X)] - \mathbb{E}[y | x])^2 \right]}_{\text{bias}} + \underbrace{\mathbb{E}_x \left[\mathbb{E}_X \left[(\mu(X) - \mathbb{E}_X [\mu(X)])^2 \right] \right]}_{\text{variance}}.$$

This exact form of bias-variance decomposition is correct for square loss in regression.

However, it is much more general.

SHAP values

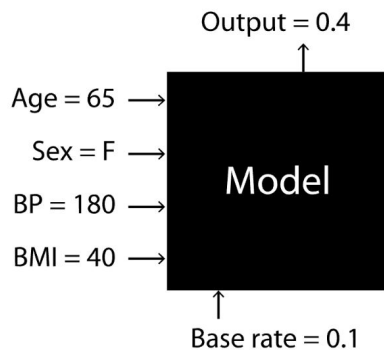
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02

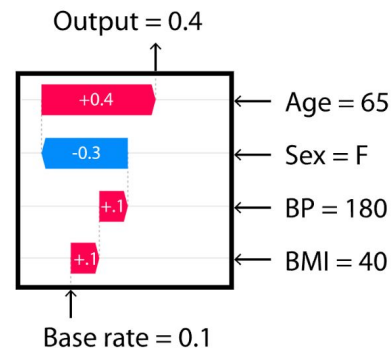
Shap values



SHAP



Explanation



<https://github.com/shap/shap>

Shap values calculation



Coalition values

$$C_{12} = 10,000$$

$$C_{12}^1 = 7,500$$

$$C_{12}^2 = 5,000$$

$$C_0 = 0$$

Marginal contribution

The increase in a coalition's value due to a player joining that coalition



$$C_{12} - C_2 = 5,000$$

$$C_{12}^1 - C_0 = 7,500$$

$$(5,000 + 7,500) / 2 = \$6,250$$



$$C_{12} - C_1 = 2,500$$

$$C_{12}^2 - C_0 = 5,000$$

$$(2,500 + 5,000) / 2 = \$3,750$$

Shap values calculation



$$C_{123} = 10,000$$

$$C_0 = 0$$

$$C_{12} = 7,500$$

$$C_{13} = 7,500$$

$$C_{23} = 5,000$$

$$C_1 = 5,000$$

$$C_2 = 5,000$$

$$C_3 = 0$$



$$C_{123} - C_{23} = 5,000$$

$$C_{12} - C_2 = 2,500$$

$$C_{13} - C_3 = 7,500$$

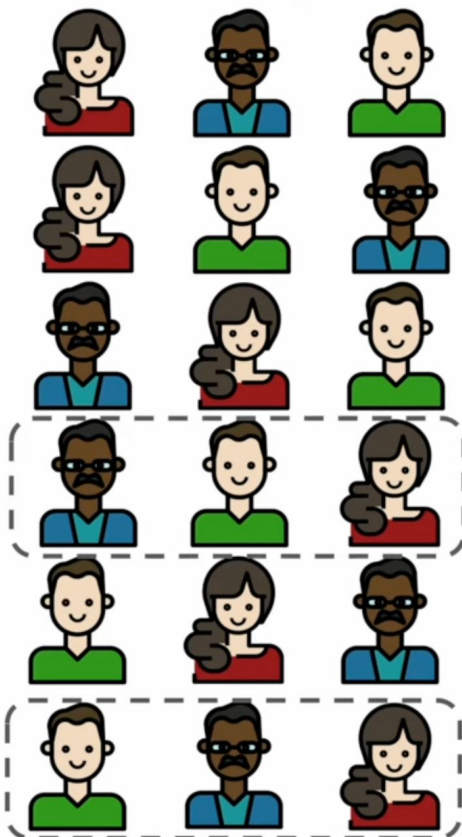
$$C_1 - C_0 = 5,000$$

$$5,000 * (1/3) + 2,500 * (1/6)$$

$$+ 7,500 * (1/6) + 5,000 * (1/3)$$

$$= \$5,000$$

Shap values calculation



$$C_{123} - C_{23} = 5,000$$

$P(C_{123} - C_{23})$ = probability that player 1 makes a marginal contribution to a coalition of player 2 and 3

$$3! = 6$$

$$2! * 1! = 2$$

$$P(C_{123} - C_{23}) = 2/6 = 1/3$$

Shap values axioms



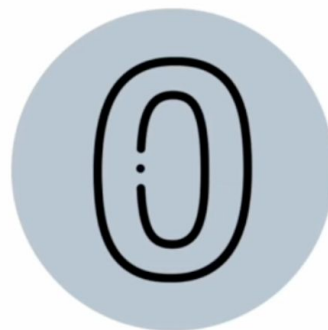
Efficiency



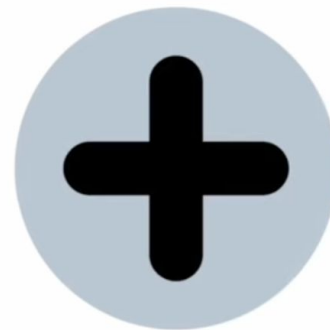
Symmetry



Null player



Additivity





Shap values

$$\phi_i = \sum_{S \subseteq \{1, \dots, p\} \setminus \{i\}} \underbrace{\frac{|S|!(p-|S|-1)!}{p!}}_{\text{Weight}} \underbrace{[val(S \cup \{i\}) - val(S)]}_{\text{Marginal contribution of player } i \text{ to coalition } S}$$

$p!$ = number of ways to form a coalition of p players

$|S|$ = number of players in coalition S

$|S|!$ = number of ways coalition S can form

$(p-|S|-1)!$ = number of ways players can join after player i joins

Shap values for ML



$$\hat{\phi}_i = \frac{1}{M} \sum_{m=1}^M (f(x_{+i}^m) - f(x_{-i}^m))$$

Diagram illustrating the calculation of Shap values for feature x_i using the game-theoretic approach. The diagram shows a grid of feature combinations (represented by colored circles) for features x_1, x_2, x_3, x_4, x_5 . The top row shows the full set of features. Below it, a horizontal line separates the full set from the combinations used in the Shap value calculation. The grid shows combinations where one feature is present (colored) and the others are absent (gray), and vice versa. The Shap value for feature x_i is calculated as the average difference in the model output f between the combinations where x_i is present and where it is absent, across M samples.

Shap values for ML



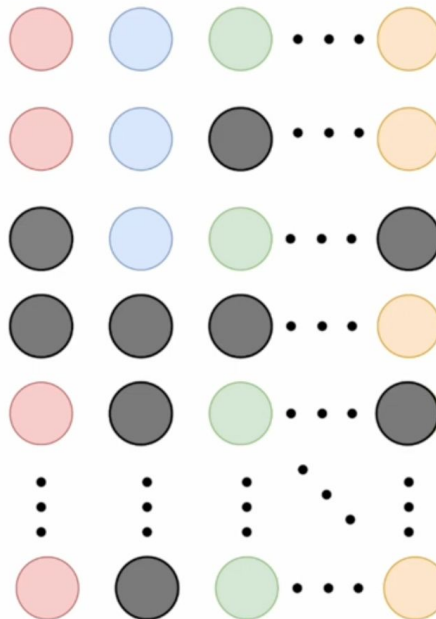
KernelSHAP



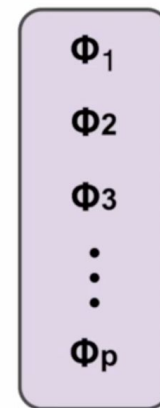
TreeSHAP



Φ_1 Φ_2 Φ_3 $\dots$ Φ_p



Linear
Regression



Libraries for shap values



- <https://shap.readthedocs.io/en/latest/index.html>
 - <https://github.com/shap/shap>
- <https://github.com/mmschlk/shapiq>
-

Materials on shap values



- shap values introduction
<https://www.youtube.com/watch?v=MQ6fFDwjuco&list=PLqDyyww9y-1SJgMw92x90qPYpHgahDLIK>
- detailed shapley values <https://www.youtube.com/watch?v=4RkhsIz14Yc>
- Explainable AI
 - https://www.youtube.com/watch?v=VY7SCI_DFho&list=PLqDyyww9y-1SwNZ-6CmvfXDAOdLS7yUQ4
 - <https://www.youtube.com/watch?v=OZJ1lgSgP9E&list=PLV8yxwGOxwovp-j6ztxhF3QcKXT6vORU>
- [ru] <https://t.me/smalldatascience/1019>
 - https://vkvideo.ru/video-232735712_456239019

Revise



1. Bias Variance Decomposition
2. SHAP Values

Thanks for attention!

Questions?



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Bias-variance decomposition derivation

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Bias-variance decomposition

The dataset $X = (x_i, y_i)_{i=1}^{\ell}$ with $y_i \in \mathbb{R}$

for regression problem.

Denote loss function $L(y, a) = (y - a(x))^2$.

The empirical risk takes form:

$$R(a) = \mathbb{E}_{x,y} \left[(y - a(x))^2 \right] = \int_{\mathbb{X}} \int_{\mathbb{Y}} p(x, y) (y - a(x))^2 dx dy.$$

Bias-variance decomposition



Let's show that

$$a_*(x) = \mathbb{E}[y | x] = \int_{\mathbb{Y}} yp(y | x)dy = \arg \min_a R(a).$$

$$\begin{aligned} L(y, a(x)) &= (y - a(x))^2 = (y - \mathbb{E}(y | x) + \mathbb{E}(y | x) - a(x))^2 = \\ &= (y - \mathbb{E}(y | x))^2 + 2(y - \mathbb{E}(y | x))(\mathbb{E}(y | x) - a(x)) + (\mathbb{E}(y | x) - a(x))^2. \end{aligned}$$

Returning to the risk estimation:

$$\begin{aligned} R(a) &= \mathbb{E}_{x,y} L(y, a(x)) = \\ &= \mathbb{E}_{x,y} (y - \mathbb{E}(y | x))^2 + \mathbb{E}_{x,y} (\mathbb{E}(y | x) - a(x))^2 + \\ &\quad + 2\mathbb{E}_{x,y} (y - \mathbb{E}(y | x)) (\mathbb{E}(y | x) - a(x)). \end{aligned}$$



$$R(a) = \mathbb{E}_{x,y} L(y, a(x)) =$$

Focus on the last term:

$$= \mathbb{E}_{x,y} (y - \mathbb{E}(y | x))^2 + \mathbb{E}_{x,y} (\mathbb{E}(y | x) - a(x))^2 + 2\mathbb{E}_{x,y} (y - \mathbb{E}(y | x)) (\mathbb{E}(y | x) - a(x)).$$

Does not depend on y

$$\begin{aligned} \mathbb{E}_x \mathbb{E}_y \left[(y - \mathbb{E}(y | x)) (\mathbb{E}(y | x) - a(x)) \mid x \right] &= \\ &= \mathbb{E}_x \left((\mathbb{E}(y | x) - a(x)) \mathbb{E}_y \left[(y - \mathbb{E}(y | x)) \mid x \right] \right) = \\ &= \mathbb{E}_x \left((\mathbb{E}(y | x) - a(x)) (\mathbb{E}(y | x) - \mathbb{E}(y | x)) \right) = \\ &= 0 \end{aligned}$$



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So the risk takes form:

$$R(a) = \mathbb{E}_{x,y}(y - \mathbb{E}(y | x))^2 + \mathbb{E}_{x,y}(\mathbb{E}(y | x) - a(x))^2.$$

Does not depend on $a(x)$

The minimum is reached when $a(x) = \mathbb{E}(y | x).$

So the optimal regression model with square loss is

$$a_*(x) = \mathbb{E}(y | x) = \int_{\mathbb{Y}} yp(y | x)dy.$$

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In further slides (x) is omitted!



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If X is fixed, then

$$\mathbb{E}_{x,y} \left[(y - \mu(X))^2 \right] = \mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right] + \mathbb{E}_{x,y} \left[(\mathbb{E}[y | x] - \mu(X))^2 \right].$$



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$$L(\mu) = \mathbb{E}_X \left[\underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{Does not depend on X}} + \mathbb{E}_{x,y} \left[(\mathbb{E}[y | x] - \mu(X))^2 \right] \right]$$

Does not depend on X

$$L(\mu) = \mathbb{E}_X \left[\underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{Does not depend on X}} + \mathbb{E}_{x,y} \left[(\mathbb{E}[y | x] - \mu(X))^2 \right] \right] =$$



$$L(\mu) = \mathbb{E}_X \left[\underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{Does not depend on } X} + \mathbb{E}_{x,y} \left[(\mathbb{E}[y | x] - \mu(X))^2 \right] \right] =$$



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Focus on the second term:

$$L(\mu) = \mathbb{E}_X \left[\underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]} + \mathbb{E}_{x,y} \left[(\mathbb{E}[y | x] - \mu(X))^2 \right] \right] =$$



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$$\begin{aligned}\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mu(X) \right)^2 \right] \right] &= \\ &= \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] + \mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] =\end{aligned}$$





$$\begin{aligned}\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mu(X) \right)^2 \right] \right] &= \\ &= \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] + \mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] = \\ &= \mathbb{E}_{x,y} \left[\underbrace{\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right)^2 \right]}_{\text{var}(\mathbb{E}[y | x])} \right] + \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] + \\ &\quad + 2 \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right) \left(\mathbb{E}_X [\mu(X)] - \mu(X) \right) \right] \right].\end{aligned}$$



$$\begin{aligned}\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mu(X) \right)^2 \right] \right] &= \\&= \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] + \mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] = \\&= \mathbb{E}_{x,y} \left[\underbrace{\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right)^2 \right]}_{\text{Does not depend on } X} \right] + \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] + \\&\quad + 2 \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right) \left(\mathbb{E}_X [\mu(X)] - \mu(X) \right) \right] \right].\end{aligned}$$



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Just a bit further, we are almost there



$$\begin{aligned}\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mu(X) \right)^2 \right] \right] &= \\ &= \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] + \mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] = \\ &= \mathbb{E}_{x,y} \left[\underbrace{\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right)^2 \right]} + \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}_X [\mu(X)] - \mu(X) \right)^2 \right] \right] + \right. \\ &\quad \left. + 2 \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[\left(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)] \right) \left(\mathbb{E}_X [\mu(X)] - \mu(X) \right) \right] \right] \right].\end{aligned}$$

Focus on this term

$$\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)]) (\mathbb{E}_X [\mu(X)] - \mu(X)) \right] =$$



$$\begin{aligned}\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)]) (\mathbb{E}_X [\mu(X)] - \mu(X)) \right] &= \\ &= (\mathbb{E}[y | x] - \mathbb{E}_X [\mu(X)]) \mathbb{E}_X [\mathbb{E}_X [\mu(X)] - \mu(X)] =\end{aligned}$$



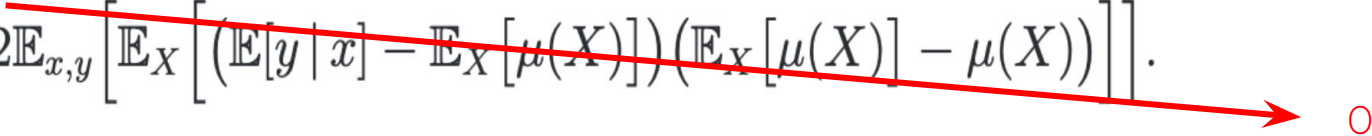
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$$\begin{aligned}\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mu(X))^2 \right] \right] &= \\&= \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mathbb{E}_X[\mu(X)] + \mathbb{E}_X[\mu(X)] - \mu(X))^2 \right] \right] = \\&= \mathbb{E}_{x,y} \left[\underbrace{\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mathbb{E}_X[\mu(X)])^2 \right]}_{\text{var}(\mathbb{E}[y | x])} \right] + \mathbb{E}_{x,y} \left[\mathbb{E}_X \left[(\mathbb{E}_X[\mu(X)] - \mu(X))^2 \right] \right] + \\&\quad + 2\mathbb{E}_{x,y} \left[\mathbb{E}_X \left[(\mathbb{E}[y | x] - \mathbb{E}_X[\mu(X)])(\mathbb{E}_X[\mu(X)] - \mu(X)) \right] \right].\end{aligned}$$




$$L(\mu) = \underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{noise}} +$$
$$+ \underbrace{\mathbb{E}_x \left[(\mathbb{E}_X [\mu(X)] - \mathbb{E}[y | x])^2 \right]}_{\text{bias}} + \underbrace{\mathbb{E}_x \left[\mathbb{E}_X \left[(\mu(X) - \mathbb{E}_X [\mu(X)])^2 \right] \right]}_{\text{variance}}.$$



$$L(\mu) = \underbrace{\mathbb{E}_{x,y} \left[(y - \mathbb{E}[y | x])^2 \right]}_{\text{noise}} + \underbrace{\mathbb{E}_x \left[(\mathbb{E}_X [\mu(X)] - \mathbb{E}[y | x])^2 \right]}_{\text{bias}} + \underbrace{\mathbb{E}_x \left[\mathbb{E}_X \left[(\mu(X) - \mathbb{E}_X [\mu(X)])^2 \right] \right]}_{\text{variance}}.$$

This exact form of bias-variance decomposition is correct for square loss in regression.

However, it is much more general. See extra materials for more exotic cases.

